

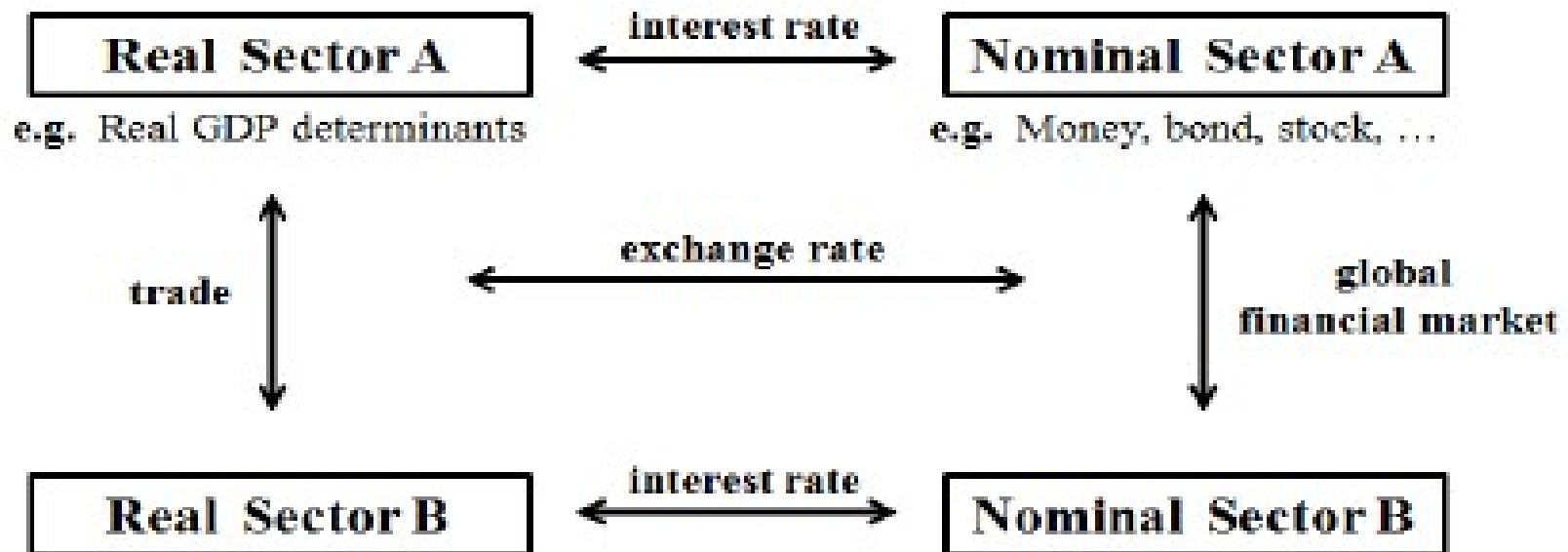
국제 금융

기본 모델

□ 2국가 모델

- Two countries, A and B
- Each economy comprises of nominal and real sectors
- Strong interactions among the sectors across countries

Figure 1.1: Two Sectors of Economy



기본 모델

□ 2국가 모델

- Key questions

- Why exchange rates matter?
- How to forecast exchange rates?

Arbitrage-free Exchange Rate

Arbitrage-free Exchange Rate

- Partial Equilibrium Approach
- Exchange rate: price 1 USD in terms of KRW

Absolute PPP

□ 절대 구매력 평가설

- Assumptions

1. Two countries (KOR, US)
2. All goods are identical and tradable
3. No transaction cost
4. CPI weights are the same across the countries

- Notations

- P_i = unit price of good i in terms of KRW
- P_i^* = unit price of good i in terms of USD
- S = unit price of USD in terms of KRW (*exchange rate*)
- W_i = CPI weight of good i
- P = CPI of KOR
- P^* = CPI of US

Absolute PPP

□ 절대 구매력 평가설

■ Derivation

- By assumptions 1, 2, and 3, we have

$$P_i = P_i^* \times S$$

for $\forall_i = 1, 2, \dots, N$. (P_i^* = number of USD you need, S = price of 1 USD in terms of KRW)

- By assumption 4,

$$\begin{array}{r} W_1 P_1 = W_1 P_1^* S \\ W_2 P_2 = W_2 P_2^* S \\ \vdots \\ W_N P_N = W_N P_N^* S \\ \hline P = P^* S \end{array}$$

Absolute PPP

□ 절대 구매력 평가설

- Finally, the nominal exchange rate (S) is determined as

$$P = P^* S \Leftrightarrow S = \frac{P}{P^*} \quad \text{under assumptions a,b,c, and d} \quad (3.1)$$

⇒ one equation, three variables $\therefore$ two of them should be *exogenous*

⇒ given P and P^* , S is an endogenous variable

⇒ **partial equilibrium approach**

- The prices, P and P^* are exogenously determined and how they are generated is unknown.
- **Limitation:** Absolute PPP does not hold empirically because the assumptions are unrealistic

Relative PPP

□ 상대 구매력 평가설

- It is often found that in some particular countries, there exists a constant, k such that

$$P_t = kP_t^*S_t.$$

where P_t, P_t^* , and S_t are time-varying. For instance, if $P_t > P_t^*S_t$, then $k > 1$. This friction is adjusted by the constant, k

- It does not have a solid theoretical foundation, just holds empirically.

$$\begin{array}{rcl} P_t & = & kP_t^*S_t \\ -\log & & \\ P_{t+1} & = & kP_{t+1}^*S_{t+1} \\ \hline \log P_{t+1} - \log P_t & = & \log P_{t+1}^* - \log P_t^* + \log S_{t+1} - \log S_t \\ \underbrace{\log P_{t+1} - \log P_t}_{\pi_{t+1}} & = & \underbrace{\log P_{t+1}^* - \log P_t^*}_{\pi_{t+1}^*} + \underbrace{\log S_{t+1} - \log S_t}_{s_{t+1}} \end{array}$$

$$\pi_{t+1} = \pi_{t+1}^* + s_{t+1} \quad (3.2)$$

where s_{t+1} is a depreciation rate

Uncovered Interest Rate Parity

□ 위험 이자율 평형설 (UIP)

- Assumptions

- a No transaction cost (perfect capital mobility)

- b Risk-neutral agents

- Derivation: Suppose that an investor has 1 KRW and two investment options

- (i) Domestic investment (risk-free)

- : payoff after 1 year = $1 \times (1 + i_t)$

- (ii) Foreign investment (risky, because exchange rate is uncertain)

- : payoff after 1 year = $1 \times \frac{1}{S_t} \times (1 + i_t^*) \times S_{t+1}^e$

⇒ Due to the assumption b, options (i) and (ii) are indifferent to the investor, which leads to the UIP equation as

$$S_t = \frac{1 + i_t^*}{1 + i_t} \times S_{t+1}^e \quad (3.3)$$

Uncovered Interest Rate Parity

□ 위험 이자율 평형설 (UIP)

- Meaning of S_{t+1}^e : Take log on both sides of the UIP equation as

$$S_t = \frac{1 + i_t^*}{1 + i_t} \times S_{t+1}^e$$

$$\Rightarrow \log S_t = i_t^* - i_t + \log S_{t+1}^e$$

Similarly, we can have

$$\log S_{t+1} = i_{t+1}^* - i_{t+1} + \log S_{t+2}^e.$$

Uncovered Interest Rate Parity

□ 위험 이자율 평형설 (UIP)

- Now suppose that we are at time t . As the UIP holds at each time point,

$$\begin{aligned}\log S_t &= i_t^* - i_t + \log S_{t+1}^e \\ \log S_{t+1}^e &= i_{t+1}^{*e} - i_{t+1}^e + \log S_{t+2}^e \\ \log S_{t+2}^e &= i_{t+2}^{*e} - i_{t+2}^e + \log S_{t+3}^e\end{aligned}$$

$$+ \quad \vdots \quad \quad \quad \vdots$$

$$\underbrace{\log S_t}_{\text{today's exchange rate}} = \underbrace{(i_t^* - i_t) + (i_{t+1}^{*e} - i_{t+1}^e) + (i_{t+2}^{*e} - i_{t+2}^e) + \dots}_{\text{stream of interest rate differentials across countries}}$$

$\Rightarrow$ even if today's interest rate does not change, as long as future interest rate expectations change, today's exchange rate changes.

$$S_t = \underbrace{\frac{1 + i_t^*}{1 + i_t}}_{\text{today's interest rate}} \times \underbrace{S_{t+1}^e}_{\text{information about future interest rates}}$$

Uncovered Interest Rate Parity

□ UIP의 실증 테스트

- Empirical test of the UIP: The UIP implies

$$\log S_t = i_t^* - i_t + \log S_{t+1}^e$$

and

$$\begin{aligned}\log S_{t+1}^e - \log S_t &= \text{expected depreciation rate} \\ &= i_t - i_t^*\end{aligned}$$

- If the expected exchange rate, S_{t+1}^e is replaced by its realization, S_{t+1} , then

$$s_{t+1} = \underbrace{\frac{S_{t+1} - S_t}{S_t}}_{\text{depreciation rate}} = i_t - i_t^* + \underbrace{\text{error}}_{\because S_{t+1}^e \neq S_{t+1}}$$

Uncovered Interest Rate Parity

□ UIP의 실증 테스트

- Regression analysis: $s_{t+1} = \frac{S_{t+1} - S_t}{S_t} = \beta_1 + \beta_2(i_t - i_t^*) + error$
 - ⇒ If $\beta_1 = 0, \beta_2 = 1$, then UIP holds
 - ⇒ In reality, $\hat{\beta}_1 \neq 0$ and $\hat{\beta}_2 \ll 1$ (around 0, or sometimes negative)
- Why UIP does not hold empirically?
 - ⇒ Perhaps it is because the assumption "risk-neutral agents" is unrealistic. The assumption "no transaction cost" is plausible, too.

Covered Interest Rate Parity

□ 무위험 이자율 평형설 (CIP)

• Assumptions

- No transaction cost (perfect capital mobility)
- No default risk
- Risk-neutral preference is no longer required

• Forward exchange rate

$F_{t,t+1}$: contract on t , delivery on $t + 1 \Rightarrow$ no uncertainty, and no risk

Example a contract to change 1 USD for 1,000 KRW after a year

$$\Rightarrow F_{t,t+1} = 1,000 \text{ KRW}$$

$$\underbrace{1 + i_t}_{\text{payoff of domestic investment with 1KRW}} = \underbrace{\frac{1}{S_t}(1 + i_t^*)F_{t,t+1}}_{\text{payoff of foreign investment}} \quad (3.4)$$

Since there is no expectation term, nothing is random $\Rightarrow$ risk-free

Covered Interest Rate Parity

□ 무위험 이자율 평형설 (CIP)

• CIP equation

$$\underbrace{1 + i_t}_{\text{payoff of domestic investment with 1KRW}} = \underbrace{\frac{1}{S_t}(1 + i_t^*)F_{t,t+1}}_{\text{payoff of foreign investment}} \quad (3.5)$$

$\Rightarrow$

$$\text{CIP : } \underbrace{F_{t,t+1}}_{\text{endogenous}} = \frac{1 + i_t}{1 + i_t^*} \times \underbrace{S_t}_{\text{exogenous}} \quad (3.6)$$

• Difference from UIP

- In UIP S_t is an endogenous variable
- In CIP $F_{t,t+1}$ is endogenous whereas S_t is exogenous

Covered Interest Rate Parity

□ 실증 분석

- Empirical analysis

- does not hold usually empirically, while CIP does.
- If there is a default risk like in a forward contract between developing and developed countries, CIP does not hold.

Forward Premium Puzzle

□ Forward Premium Puzzle

- Forward premium

– Definition: $\frac{F_{t,t+1} - S_t}{S_t} = \log F_{t,t+1} - \log S_t$

$$(\because \log \frac{F_{t,t+1}}{S_t} = \log \frac{F_{t,t+1} - S_t + S_t}{S_t} = \log(1 + \frac{F_{t,t+1} - S_t}{S_t}) \approx \frac{F_{t,t+1} - S_t}{S_t})$$

According to CIP, $\log F_{t,t+1} - \log S_t = i_t - i_t^*$

Forward Premium Puzzle

□ Forward Premium Puzzle

- Forward premium puzzle

- From the regression equation $s_{t+1} = \beta_1 + \beta_2 \underbrace{(i_t - i_t^*)}_{\text{forward premium}} + \text{error}$, the coefficients are estimated as $\beta_1 \neq 0$ and $\beta_2 \ll 1$
- The reason why UIP does not hold is not clear (may be because of home bias, regulations, psychological impacts, risk-neutral agent assumption, etc.)
- Suppose UIP and CIP hold in reality:

$$S_t = \frac{1 + i_t^*}{1 + i_t} S_{t+1}^e$$

and

$$S_t = \frac{1 + i_t^*}{1 + i_t} F_{t,t+1}.$$

Then, $S_{t+1}^e = F_{t,t+1}$.

Forward Premium Puzzle

□ Forward Premium Puzzle

- If you want to obtain an expected future exchange rate (i.e., S_{t+1}^e = market expectation), you can just look up the forward exchange rate which can be known from the forward market.
- *However, UIP does not hold in reality.* That is, $S_{t+1}^e \neq F_{t,t+1}$. Therefore, $F_{t,t+1}$ is **not** a good proxy of S_{t+1}^e .

Forward Premium Puzzle

□ Forward Premium Puzzle

[Note] Consider the following statement: the UIP equation,

$$S_t = \frac{1 + i_t^*}{1 + i_t} S_{t+1}^e$$

indicates that i_t rises by 1% in response to the rise of i_t^* by 1%

- This statement is not quite correct. It is because S_t is the only endogenous variable
- Since it is a partial equilibrium, only S_t fluctuates and other variables should be exogenously given. Exogenous variables cannot be affected by other exogenous variables. If they do, another structural equations are needed
- In other words, we do not know how i_t is determined, so it is not sensible to say that i_t goes up just because i_t^* goes up

Fisher Equation

□ 피셔 방정식

• Assumptions

- (a) risk-neutral agents
- (b) no arbitrage condition
- (c) The real return is non-random i.e., the real return is known at present ($r_{t+1}^e = r_t$)

• Derivation

- K_t : quantity of real goods (measured in real variables)
- There are two investment options with K_t
 - (i) sell K_t at price P_t
 - $\Rightarrow P_t K_t$ cash $\Rightarrow$ deposit $\Rightarrow$ 1 year later, $(1 + i_t)P_t K_t$ payoff
 - (ii) keep K_t for a year
 - $\Rightarrow$ at time $t + 1$, K_{t+1} (productivity changes) at P_{t+1} and sell it
 - $\Rightarrow$ 1 year later, $K_{t+1}P_{t+1}$ payoff

Fisher Equation

□ 피셔 방정식

$$\begin{aligned}\therefore \underbrace{P_t K_t (1 + i_t)}_{\text{risk-free}} &= \underbrace{P_{t+1}^e K_{t+1}^e}_{\text{risky}} \\ \Rightarrow \underbrace{1 + i_t}_{\text{nominal returns}} &= \frac{P_{t+1}^e}{P_t} \underbrace{\frac{K_{t+1}^e}{K_t}}_{\text{real returns}}\end{aligned}$$

$\Rightarrow$ take log to have

$$i_t = \pi_{t+1}^e + r_{t+1}^e = \pi_{t+1}^e + r_t.$$

By the assumption that $r_{t+1}^e = r_t$, it follows that

$$i_t = \pi_{t+1}^e + r_t : \text{Fischer Equation} \quad (3.7)$$

Relation among PPP, UIP and Fisher Equation

□ PPP, UIP와 Fisher 방정식

- Assumptions

- (a) real interest rates are the same across countries : $r_t = r_t^*$
- (b) no uncertainty

- Two of the followings imply the other

- (a) Relative PPP: $s_t = \pi_t - \pi_t^*$
- (b) UIP: $s_{t+1} = i_t - i_t^*$
- (c) Fisher Equation: $i_t = r_t + \pi_{t+1}$

Relation among PPP, UIP and Fisher Equation

▣ 상대 PPP + 피셔 방정식 $\Rightarrow$ UIP

1. Relative PPP + Fisher equation imply UIP

As the Fischer equation holds for both countries,

$$\begin{aligned} i_t &= r_t + \pi_{t+1} \\ - \quad \underline{i_t^* &= r_t^* + \pi_{t+1}^*}. \end{aligned}$$

This yields that

$$\begin{aligned} i_t - i_t^* &= r_t - r_t^* + \pi_{t+1} - \pi_{t+1}^* \\ &= \pi_{t+1} - \pi_{t+1}^* \quad (\because r_t = r_t^*) \\ &= s_{t+1} \quad (\because PPP) \end{aligned}$$

$\therefore s_{t+1} = i_t - i_t^* : \text{UIP}$

Relation among PPP, UIP and Fisher Equation

▣ UIP + 피셔 방정식 $\Rightarrow$ 상대 PPP

2. UIP + Fisher equation imply Relative PPP

by UIP,

$$\begin{aligned}s_{t+1} &= i_t - i_t^* \\ &= r_t + \pi_{t+1} - i_t \quad (\because \text{Fisher Equation}) \\ &= \pi_{t+1} - \pi_{t+1}^* \quad (\because i_t^* = r_t^* + \pi_{t+1}^*)\end{aligned}$$

$$\therefore s_{t+1} = \pi_{t+1} - \pi_{t+1}^* : \text{PPP}$$

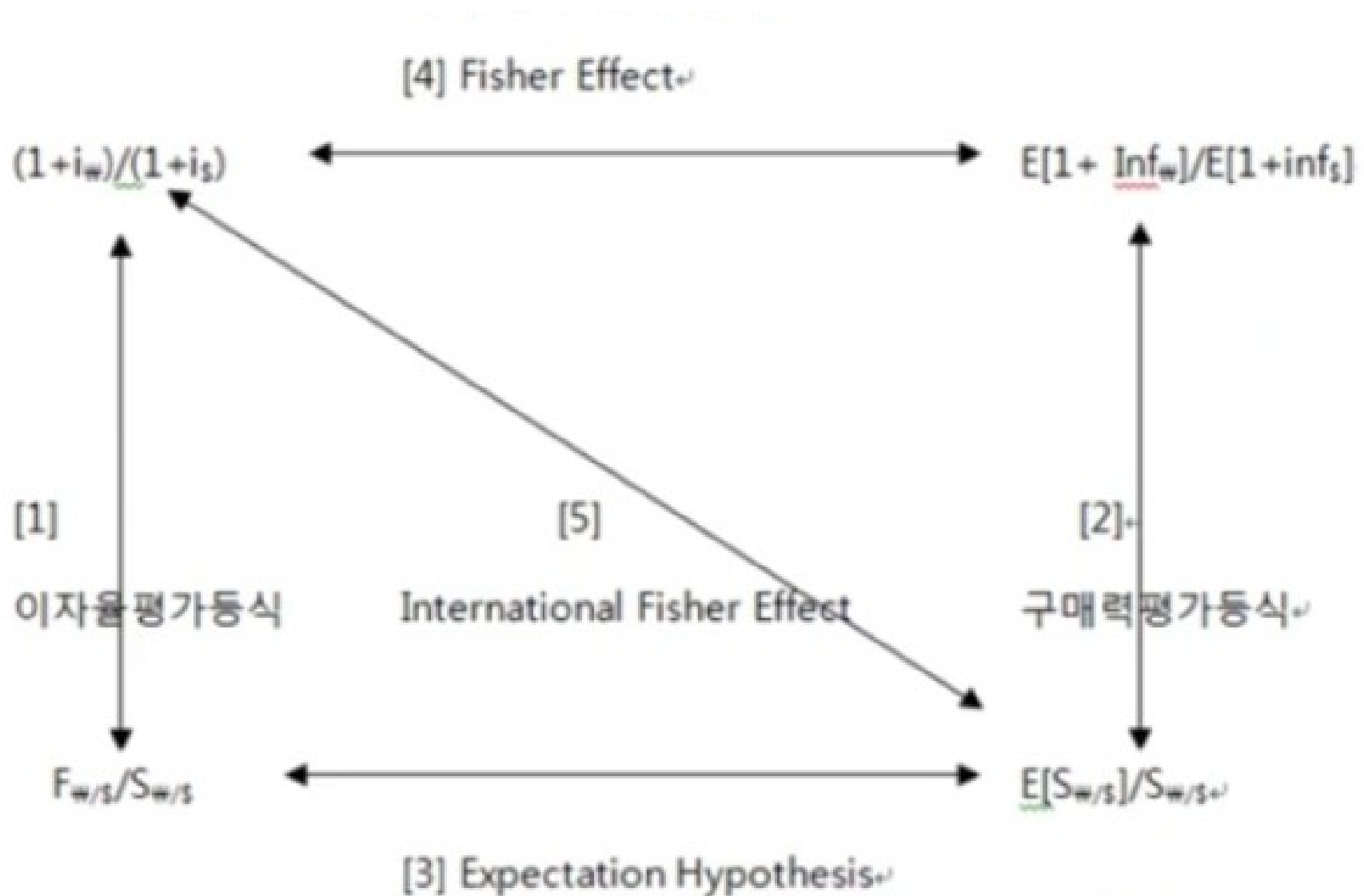
Relation among PPP, UIP and Fisher Equation

☐ $UIP + PPP \Rightarrow$ 피셔 방정식

3. $UIP + PPP$ imply the Fisher equation: Homework

종합적 관계

□ 종합적 관계



CAPM관점에서 UIP

UIP with Risk Aversion

□ 수정 SDF (확률적 할인계수)

- Problems with SDF that we covered

$$\log M_{t+1} = \log \beta - \gamma C'_{t+1} \quad (\text{from utility maximization})$$

$$\begin{aligned} E(r_{t+1}) - r_f &= -\text{Cov}(\log M_{t+1}, r_{t+1}) \\ &= \gamma \text{Cov}(C'_{t+1}, r_{t+1}) \end{aligned}$$

Problem 1 Lots of things to consider except consumption

In case of a house, other factors like population growth, inflation, etc. should be considered. In case of a stock, unemployment rate, inflation rate, etc. should be taken into account.

Problem 2 Poor prediction

Problem 3 Unknown utility function

UIP with Risk Aversion

□ 수정 SDF (확률적 할인계수)

- **Modified SDF by finance:** Less solid theoretical foundation

$$\log M_{t+1} = -r_f - \frac{1}{2}\gamma^2 - \gamma e_{t+1}, \quad (3.15)$$

$$e_{t+1} \sim i.i.d\mathcal{N}(0, 1) \quad (3.16)$$

- C_{t+1} is replaced by e_{t+1}
- e_{t+1} can be multivariate, anything that help predict future returns
- This SDF enables us to forecast future return more accurately $\therefore$ used in general
- The term, $\frac{1}{2}\gamma^2$ is required for the SDF to satisfy the condition that the expectation of the SDF is equal to the risk-free asset price:

$$P_t = \mathbb{E}(M_{t+1} \cdot P_{t+1}) = \mathbb{E}(M_{t+1}) = \frac{1}{1 + r_f}$$

- γ measures the degree of risk-aversion. The higher value of γ indicates the more risk-averse

UIP with Risk Aversion

□ UIP with Risk Aversion

1. Assumptions

- (a) Two countries
- (b) SDF is unique for each country

2. Pricing formulas

- $P_t = \mathbb{E}(M_{t+1} \cdot P_{t+1})$: pricing formula for domestic assets in KRW
- $P_t^* = \mathbb{E}(M_{t+1}^* \cdot P_{t+1}^*)$: price formula for foreign assets in USD

UIP with Risk Aversion

□ UIP with Risk Aversion

3. Derivation

- Consider a risky foreign asset such that its payoff at $t + 1$ is randomly determined as

$$P_{t+1}^* = \frac{1}{S_{t+1}} \quad (3.17)$$

- Note that $P_{t+1}^* = 1/S_{t+1}$ is in USD
- Now consider a risk-free domestic asset such that

$$P_t = \mathbb{E}(M_{t+1} \cdot 1), \quad (3.18)$$

where the payoff is fixed at one

- By the CAPM, the price of the foreign asset is

$$P_t^* = \mathbb{E} \left(M_{t+1}^* \cdot \frac{1}{S_{t+1}} \right)$$

, which can be rewritten as

$$S_t P_t^* = \mathbb{E} \left(M_{t+1}^* \cdot \frac{S_t}{S_{t+1}} \right)$$

This is the price of the foreign asset in KRW, not in USD

UIP with Risk Aversion

□ UIP with Risk Aversion

- Here notice that the expected payoff of the foreign asset is one KRW without uncertainty. That is,

$$S_t P_t^* = \mathbb{E} \left(M_{t+1}^* \cdot \frac{S_t}{S_{t+1}} \cdot \left(\frac{1}{S_{t+1}} S_{t+1} \right) \right)$$

- $(\frac{1}{S_{t+1}}, \text{payoff in USD}) \times (S_{t+1}, \text{exchange rate}) = \text{one KRW}$. This is non-random although $\frac{1}{S_{t+1}}$ is random.
- By the LOP, the prices of the domestic and foreign assets in KRW should be the same because the assets are identical:

$$P_t = S_t P_t^*$$

and

$$\mathbb{E}(M_{t+1} \cdot 1) = \mathbb{E} \left(\underbrace{M_{t+1}^* \cdot \frac{S_t}{S_{t+1}}}_{\text{domestic SDF}} \cdot 1 \right)$$

UIP with Risk Aversion

□ UIP with Risk Aversion

- By the assumption of the unique country-specific SDF, we have

$$M_{t+1} = M_{t+1}^* \cdot \frac{S_t}{S_{t+1}},$$

and in log,

$$\log M_{t+1} = \log M_{t+1}^* + \log S_t - \log S_{t+1}.$$

- The log difference of the SDFs is the depreciation rate,

$$\begin{aligned} \log S_{t+1} - \log S_t &= \log M_{t+1}^* - \log M_{t+1} \\ &= -r_f^* + r_f - \frac{1}{2}(\gamma^{*2} - \gamma^2) - \gamma^* e_{t+1}^* + \gamma e_{t+1} \end{aligned}$$

UIP with Risk Aversion

□ UIP with Risk Aversion

- Finally, the UIP equation under risk-averse agent is derived as

$$\mathbb{E}(s_{t+1}) = r_f - r_f^* + \frac{1}{2}(\gamma^2 - \gamma^{*2}) \quad (3.19)$$

because $\mathbb{E}(e_{t+1}^*) = E(e_{t+1}) = 0$

- The risk-neutral UIP is a special case of this generalized UIP when the risk-aversion coefficients are zeros ($\gamma = 0, \gamma^* = 0$)

UIP with Risk Aversion

- Exchange Rate Risk Premium

4. Exchange rate risk-premium

- ◇ Case 1: $\gamma > \gamma^*$

Suppose that $\gamma > 0, \gamma^* = 0$ for simplicity. Then,

$$s_{t+1}^e + r_f^* = r_f + \frac{1}{2}\gamma^2 > r_f$$

$s_{t+1}^e + r_f^* =$ ER from the foreign investment

$r_f =$ ER from the risk-free domestic investment

UIP with Risk Aversion

□ Exchange Rate Risk Premium

◇ Case 2: $\gamma = \gamma^*$

$$s_{t+1}^e + r_f^* = r_f$$

◇ Case 3: $\gamma < \gamma^*$

Suppose that $\gamma = 0, \gamma^* > 0$ for simplicity. Then, $s_{t+1}^e + r_f^* < r_f$

$\therefore$ the size of $\frac{1}{2}(\gamma^2 - \gamma^{*2})$ depends on the relative risk aversion across countries

UIP with Risk Aversion

□ Exchange Rate Risk Premium

[Note] We just finished the partial equilibrium approach for exchange rate determination

- advantages of partial equilibrium approach: simple, easy to derive. based on no arbitrage condition
eg. $s_{t+1}^e = i_t - i_t^*$: given i_t and i_t^* , s_{t+1}^e is determined *sequentially*.

- disadvantages of partial equilibrium approach: many variables, but only one equation and one endogenous variable

In reality, however, most variables are determined *simultaneously with strong interaction*.

∴ to make all variables endogenous, more structural equations are needed.

We now turn to a general equilibrium approach

실증 분석

UIP의 실증 테스트

□ UIP 실증 테스트

1. Prelude

$$s_{t+1}^e = \frac{S_{t+1}^e - S_t}{S_t}$$

(expected depreciation at t , so it can be different from a realized value)

2. Model

- $\therefore s_{t+1} = i_t - i_t^* + \text{prediction error (if UIP is true)}$
(under rational expectation, $\mathbb{E}(\text{prediction error})=0$)
 $\Rightarrow s_{t+1} = 1 \times \beta_1 + (i_t - i_t^*) \times \beta_2 + \text{error}_{t+1}$
- want to estimate β_1 and β_2
- s_{t+1} : dependent variable, endogenous variable
 $1, (i_t - i_t^*)$: independent variables

UIP의 실증 테스트

[Note] correlation: relation between one to one

regression: relations among one to multiple

- Among the various ways to estimate, OLS (ordinary least squares) estimation is the most popular.

$$s_{t+1} = \begin{pmatrix} 1 & i_t - i_t^* \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} + error_{t+1}$$

$$\underbrace{\begin{pmatrix} s_2 \\ s_3 \\ \vdots \\ s_T \end{pmatrix}}_Y = \underbrace{\begin{pmatrix} 1 & i_1 - i_1^* \\ 1 & i_2 - i_2^* \\ \vdots & \vdots \\ 1 & i_{T-1} - i_{T-1}^* \end{pmatrix}}_X \underbrace{\begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}}_{\beta} + \underbrace{\begin{pmatrix} e_2 \\ e_3 \\ \vdots \\ e_T \end{pmatrix}}_e$$

In a matrix form,

$$Y = X\beta + e$$

- $X\beta$ = component of Y explained by X
- e = component of Y unexplained by X

Exchange Rate Forecasting and RMSE

□ Exchange Rate Forecasting and Root Mean Squared Errors

1. Pick your independent variables you think relevant for exchange rate prediction: Suppose you choose constant term, lagged depreciation rate, domestic and foreign interest rates and inflation rates.
2. Collect the data. Then,

$$\text{Depreciation rate data} = \begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_{100} \end{pmatrix}$$

$$\text{Independent variable data} = \begin{pmatrix} 1 & s_1 & i_1 & i_1^* & \pi_1 & \pi_1^* \\ 1 & s_2 & i_2 & i_2^* & \pi_2 & \pi_2^* \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & s_{100} & i_{100} & i_{100}^* & \pi_{100} & \pi_{100}^* \end{pmatrix}$$

Exchange Rate Forecasting and RMSE

□ Exchange Rate Forecasting and Root Mean Squared Errors

3. Select important variables by the hypothesis testing. Note that

$$Y = \begin{pmatrix} s_2 \\ s_3 \\ \vdots \\ s_{100} \end{pmatrix} \text{ and } X = \begin{pmatrix} 1 & s_1 & i_1 & i_1^* & \pi_1 & \pi_1^* \\ 1 & s_2 & i_2 & i_2^* & \pi_2 & \pi_2^* \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & s_{99} & i_{99} & i_{99}^* & \pi_{99} & \pi_{99}^* \end{pmatrix}$$

4. Re-estimate the model using the selected variables (X^*) only.

5. Make a forecast. If domestic interest rate and foreign inflation rate are selected, for instance, then $\hat{s}_{101} = x_{101}^{*'} \hat{\beta}$, where

$$\hat{\beta} = (X^{*'} X^*)^{-1} X^{*'} Y$$

and

$$x_{101}^* = \begin{pmatrix} x_{1,101}^* \\ x_{2,101}^* \end{pmatrix} = \begin{pmatrix} i_{100} \\ \pi_{100}^* \end{pmatrix}.$$

Exchange Rate Forecasting and RMSE

- Exchange Rate Forecasting and Root Mean Squared Errors

6. RMSE computation

$$\text{prediction error} = s_{101} - \hat{s}_{101}$$

But, s_{101} is not available at time 101. So compute the prediction errors for the time 81, 82, ..., 100.

Next, compute the RMSE from the out-of-sample forecasting as

$$RMSE = \sqrt{\{(s_{81} - \hat{s}_{81})^2 + (s_{82} - \hat{s}_{82})^2 + \cdots + (s_{100} - \hat{s}_{100})^2\} \times \frac{1}{20}}$$

when the out-of-sample size is 20. The model producing a smaller RMSE is preferred.

7. Compare the RMSEs of your model and benchmark model. The benchmark model has the constant term only.

Currency Crisis

Currency Crisis

- 국내 통화의 급격한 가치 하락이 경기 침체를 초래할 때, 이를 설명하는 세 가지 주요 통화 위기 모델이 있다.
- 1.1세대 모델 (First-Generation Model): 이 모델은 정부의 지속 불가능한 재정 정책으로 인해 통화 위기가 발생할 수 있음을 설명한다. 정부가 자국 통화를 방어하기 위해 외환보유고를 사용하지만, 지속적인 재정적자가 외환보유고를 고갈시킨다. 결국, 통화 방어가 불가능해지면서 환율이 급등하고 통화 위기가 발생하게 된다.
- 2. 2세대 모델 (Second-Generation Model): 이 모델은 정책 결정자들이 경제 상황에 따라 고정환율제를 포기할 가능성 때문에 위기가 발생할 수 있음을 설명한다. 경기 침체나 높은 실업률과 같은 경제적 압력으로 인해 정부가 고정환율제를 유지하는데 어려움을 겪게 된다. 시장 참여자들은 정부가 고정환율제를 포기할 것이라고 예상하고, 대규모 매도가 발생하여 통화 가치가 급락한다.
- 3. 3세대 모델 (Third-Generation Model): 이 모델은 금융 부문의 부실이 통화 위기의 원인으로 작용할 수 있음을 설명한다. 기업과 은행이 외국 통화로 대규모 차입을 했을 때, 국내 통화의 급격한 평가절하가 일어나면 기업과 은행은 상환 부담이 커져 도산 위험이 높아진다. 이로 인해 금융 시스템에 불안이 생기고 자본이 유출되면서 경제 위기로 이어질 수 있다.

The First Generation Model

□ 1세대 모델

Currency crisis can happen under

(1) fiscal deficit, (2) current account deficit, and (3) fixed exchange rate
eg. Thailand (1997)

1. Current account deficit

= capital account surplus = inflow of dollars = more debt

2. Fixed exchange rate

under fixed exchange rate system, did not have enough foreign currency reserves

3. Fiscal deficit

under $S \uparrow$ pressure, expansionary fiscal policy could not be done due to fiscal deficit

4. Impossible to remain fixed exchange rate system

as foreign reserves depleted and demand for USD (to gain high profit) increased, THB depreciated drastically

The First Generation Model

□ 1세대 모델

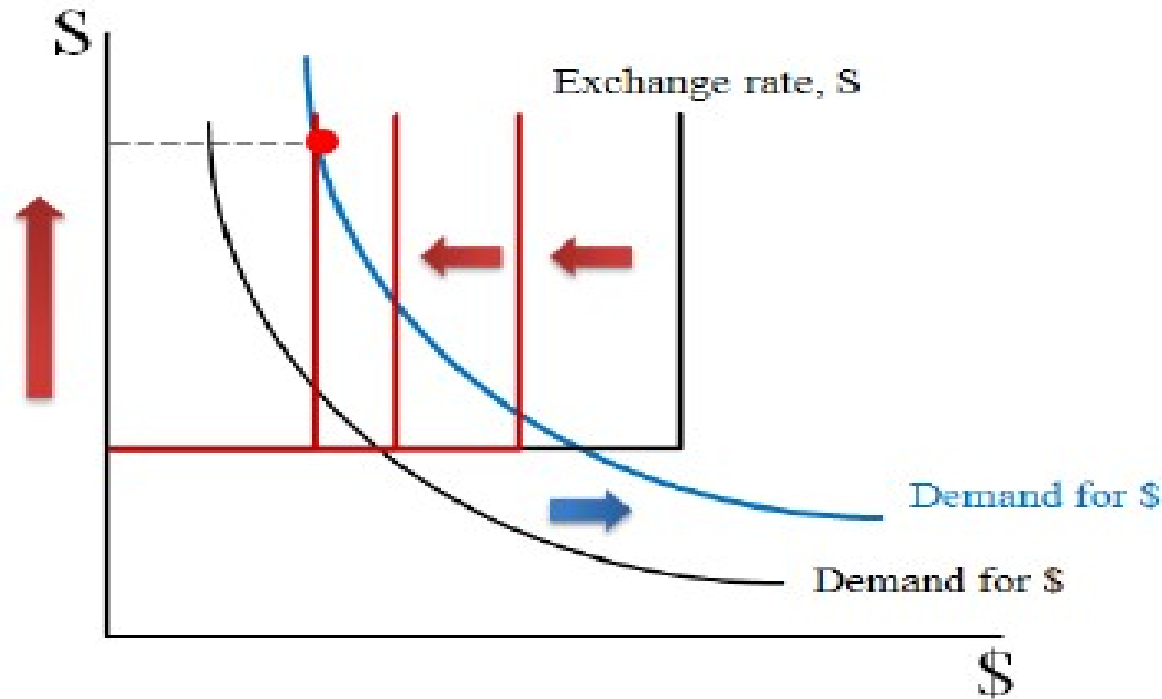


Figure 7.1: 1st generation theory

The 1st generation model could not explain the early 90's U.K crisis, so the 2nd generation model was developed. The characteristics of the currency crisis are the followings:
(1) asymmetric business cycle (2) self-fulfilling (3) crisis is a policy option
eg. The U.K crisis in the early 90's

The Second Generation Model

□ 2세대 모델

1. Fixed rate swap contract

fixed rate swap contract between the UK and Germany

∴ no need to reserve foreign currency

2. Asymmetric business cycle

high unemployment in the UK, where as high inflation in Germany

⇒ decreased supply of mark to stabilize

⇒ UK had to decrease the supply of pounds, but couldn't due to high unemployment

⇒ high depreciation pressure on pounds

3. Attack of hedge funds

hedge funds bought DEM and sold pounds

4. Crisis as a policy option

UK gave up fixed rate system even though it had an option not to give it up and this led to a currency crisis

∴ even if there's no concern about the foreign currency reserves, crisis can occur under fixed exchange rate system

The Third Generation Model

□ 3세대 모델

The 3rd generation model was developed to explain the Asian financial crisis in 1997 (under no problems on fundamentals, currency crisis break out) The characteristics of the currency crisis are the followings:

(1) contagion effect (2) balance sheet crisis
eg. Korea (1997)

1. Balance sheet problems

Suppose \$ 1 = 1,000 (actually it was \$ 1 = ₩ 800)

Asset	Liability
from (a) ₩ 1,000	from foreign bank A \$ 1 ($\because r \gg r^*$)
from (b) ₩ 1,000	from foreign bank B \$ 1
from (c) ₩ 1,000	deposit ₩ 1,000

Table 7.1: Balance Sheet of Korean Bank

2 problems on the balance sheet

- i) currency mismatch (most of liability in USD, asset in KRW)
- ii) duration mismatch (short-term liability, long-term asset)

The Third Generation Model

□ 3세대 모델

2. Contagion effect

- i) Thailand was under currency crisis,
and investors did not distinguish KOR with Thailand
- ii) Hanbo group was in bankruptcy

⇒ concern on KOR became larger ⇒ $S \uparrow$ (\$ 1 = ₩ 1,200)

3. Balance sheet crisis

liability ₩ 3,400 > asset ₩ 3,000

4. Bank run

foreign banks withdrew the short-term debts
but Korean banks cannot pay the debts back with the assets
(∵ assets were in long-term)

⇒ domestic companies (a), (b), (c) faced a liquidity problem
(∵ the debt could not be rolled over)

⇒ workers layoffs, selling firms' assets, ...

⇒ $S \uparrow$ (\$ 1 = ₩ 2,000)

Risk Management: Financial Derivatives

Risk Management: Financial Derivatives

- 환율(S)을 예측하는 것은 어렵다. 왜냐하면 거시 경제 모형(예: IS-LM-BP 모형)에 따르면 외생 변수 중 하나라도 변하면 환율이 변하기 때문에 고려해야 할 요인이 너무 많기 때문이다.
- 따라서 환율을 예측하는 것은 매우 어려운 일이다. 우리가 할 수 있는 것은 리스크를 관리하는 것이다. 리스크를 관리하는 가장 일반적인 방법은 금융 파생상품인 스왑이다.

Currency Swap

□ 통화스왑

1. Assumptions

Samsung wants to invest in the U.S. → needs 1 USD

Apple wants to invest in KOR. → needs 1,000 KRW

(at t , 1 USD = 1,000 KRW, $r = R^* = 10\%$, investment return = 20%)

2. Why it is needed

For Samsung,

(1) borrow 1,000 KRW and buy 1 USD

(2) at $t + 1$, payoff = 1.2 USD, cost = 1,000 KRW

(i) if $S = 600$ KRW at $t + 1$,

$$NPV = \frac{1.2 \times 600 - 1,100}{1.1} < 0 \text{ (still, } r = 10\%)$$

(ii) if $S = 2,000$ KRW at $t + 1$,

$$NPV = \frac{1.2 \times 2,000 - 1,100}{1.1} > 0$$

⇒ if return = 20%, NPV changes due to changes in exchange rate
(= exchange rate risk exists)

⇒ to manage the risk, use currency swap

Currency Swap

□ 통화스왑 프로세스

◇ Prerequisite for currency swap

For Samsung, borrowing cost is *higher* in US banks than in KOR banks. If the interest rates are the same across countries, currency swap does not occur. Interest rates for domestic firms are cheaper due to lower monitoring or screening costs.

Suppose,

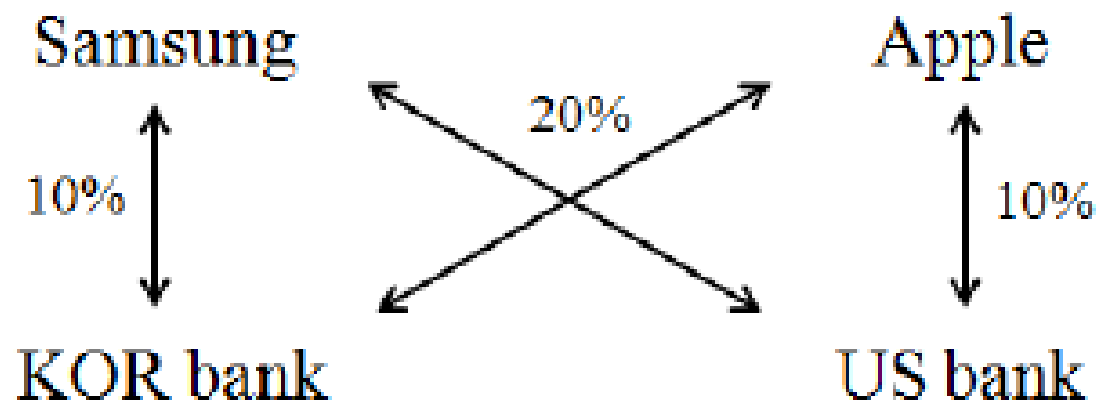


Figure 8.1: interest rates

Currency Swap

□ 통화스왑 프로세스

◇ At t ,

Samsung borrows KRW from KOR bank, and lend it to Apple.

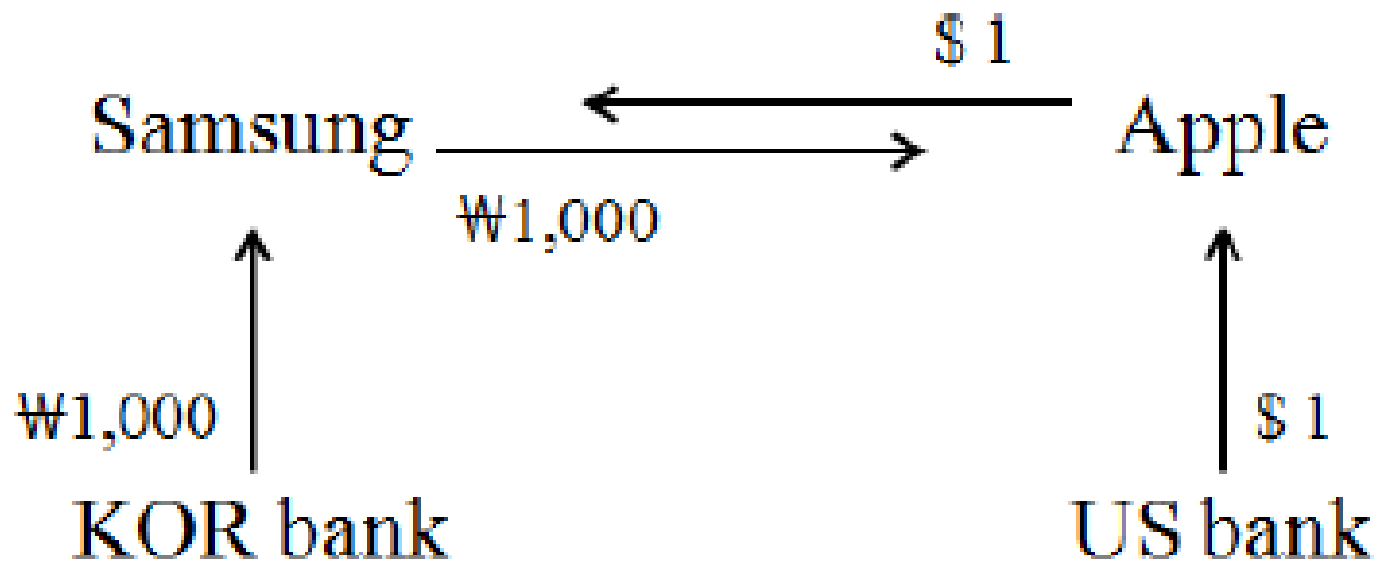


Figure 8.2: at time t transaction

Currency Swap

□ 통화스왑 프로세스

◇ At $t + 1$,

Apple pays back in KRW, and Samsung pays it back to the KOR bank.

(a) When $S = 600\text{KRW}$

$$\text{Samsung's NPV} = \frac{1.2 - 1.1}{1.1} \times 600 > 0$$

$$\text{Apple's NPV} = \frac{1200 - 1100}{1100} \times \frac{1}{600} > 0$$

Currency Swap

□ 통화스왑 프로세스

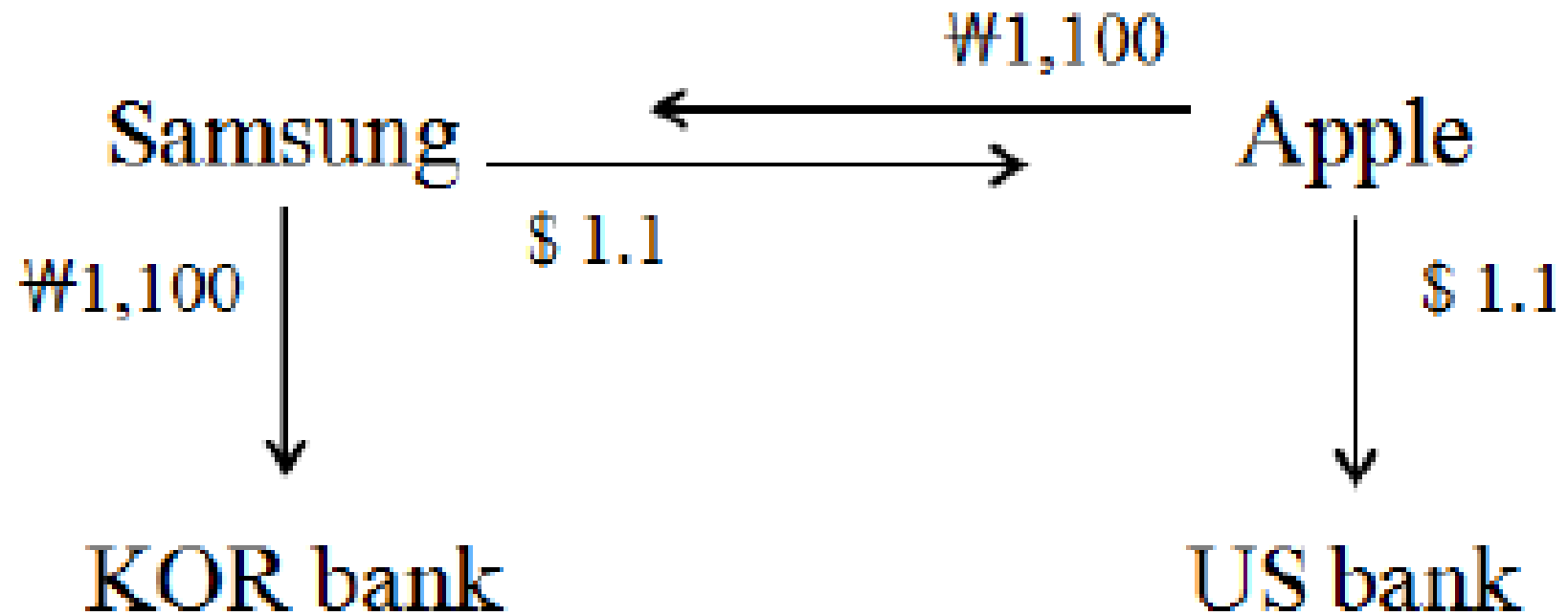


Figure 8.3: at time $t + 1$ transaction

Currency Swap

□ 통화스왑 프로세스

(b) When $S = 2,000\text{KRW}$

$$\text{Samsung's NPV} = \frac{1.2-1.1}{1.1} \times 2,000 > 0$$

$$\text{Apple's NPV} = \frac{1200-1100}{1100} \times \frac{1}{2,000} > 0$$

$\Rightarrow$ Whatever S is, a sign of NPV does not change.

$\therefore$ both firms are safer in exchange rate risks

Even though exchange rate risks disappear, counterpart risks (= default risks) still exist. Currency swap occurs because exchange rate risk is *bigger than* default risk

Interest Rate Swap

□ 이자율 스왑

1. Assumptions

	fixed interest rate	floating interest rate
big firm	10%	Libor
small firm	20%	Libor + 5%

cf. Libor: time varying short-term interest rate
(on average, $10\% > \text{Libor}$)

2. Prerequisites

(a) Big firms prefers Libor, because they have ability to endure high interest rates, but Libor is lower than fixed interest rate on average.

Small firms prefers fixed interest rates, because they fear of high unexpected interest rates.

$$(b) |20\% - 10\%| > |\text{Libor} + 5\% - \text{Libor}|$$

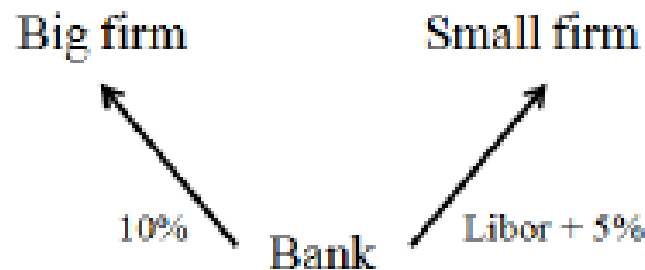
⇒ If one of these is not satisfied, swap does not occur.

Interest Rate Swap

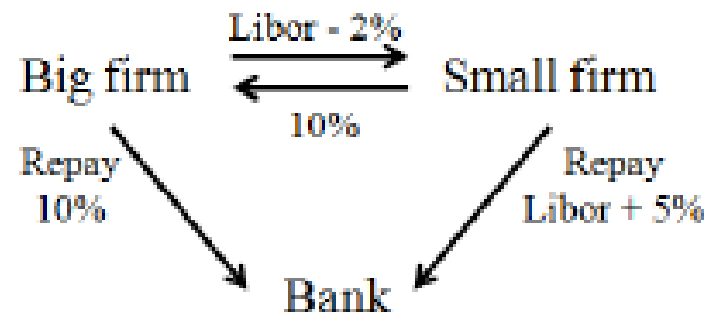
□ 이자율 스왑 프로세스

3. Process

At time t



At time $t+1$



Let's see if both firms are better off.

- Big firm: pay 10%, pay $Libor - 2\%$, paid 10%

total interest $\Rightarrow 10\% + (Libor - 2\%) - 10\% = Libor - 2\% < Libor \therefore 2\%$ benefit

- Small firm: pay $(Libor + 5\%)$, pay 10%, paid $(Libor - 2\%)$

total interest $\Rightarrow (Libor + 5\%) + 10\% - (Libor - 2\%) = 17\% < 20\% \therefore 3\%$ benefit

$\Rightarrow$ Total 5% benefit comes from differences between $|20\% - 10\%|$ and $|(Libor - 5\%) - Libor|$